AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Final Examination
Spring Semester 2025/2026

Date: 11 /Jun./2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

- Total Marks:**50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

Part 1:Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

1. Which I/O technique reduces CPU overhead for bulk data transfer?
   a) Polling.
   b) Busy Waiting.
   c) Programmed I/O.
   d) Direct Memory Access.
2. A process differs from a program because a process is:
   a) Stored on disk.
   b) Passive.
   c) An active execution.
   d) A compiled code.
3. What is the primary purpose of demand paging?
   a) To reduce the time a program takes to execute.
   b) To load all pages of a process into memory at once.
   c) To load pages only when they are needed.
   d) To manage CPU scheduling.
4. Most application programmers interact with the OS through:
   a) Assembly language.
   b) APIs.
   c) Device drivers.
   d) Interrupt vectors.
5. Which hardware feature enforces dual-mode operation?
   a) Mode Timer
   b) Cache
   c) Bus.
   d) Mode bit
6. The linker's main function is to:
   a) Combine object files.
   b) Execute programs.
   c) Load programs into memory.
   d) Both choices, a and b.
7. Which system call creates a new process in UNIX?
   a) create().
   b) exec().
   c) child().
   d) fork().
8. If a system is in the unsafe mode, this means that we have:
   a) Deadlock
   b) A possibility of a deadlock
   c) No deadlock
   d) Either a or b
9. Message passing is especially useful in:
   a) Single-user systems.
   b) Shared-memory systems.
   c) Distributed systems.
   d) Batch systems.
10. Thrashing occurs when a process:
    a) Uses minimal CPU time.
    b) Has too many frames.
    c) Executes without I/O.
    d) Spends most time paging.

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Part 2: Short and Essay questions: [15 marks]

Question 2: [3 marks]

a) Describe **system calls** and their purpose. (1.5 marks)
b) Why are **threads** called lightweight processes? (1.5 marks)

Question 3: [4 marks]

Explain the difference between **preemptive** and **non-preemptive** scheduling.

Question 4: [4 marks]

Memory management systems may use **paging** or **segmentation**.

a) Explain the difference between **internal** and **external fragmentation**. (2 marks)

b) Identify which type of fragmentation occurs in **paging** and which occurs in **segmentation** and justify your answer. (2 marks)

Question 5: [4 marks]

Answer the following questions:

a- What is the purpose of **CPU scheduling**? (2 marks)

b- Define **global replacement** in virtual memory management and list one advantage. (2 marks)

Part 3: Problem Solving questions: [25 marks]

Question 6: [6 marks]

Semaphores can be implemented without busy waiting using a **waiting queue** with each semaphore.

Describe the role of the **waiting queue** and explain the purpose of the **block** and **wakeup** operations.

Question 7: [6 marks]

Calculate the **Effective Access Time (EAT)** in **nanoseconds and milliseconds** for a demand paging system with the following parameters:

- Memory access time = **120 nanoseconds**
- Average page-fault service time = **8 milliseconds**
- Page-fault rate, $\boldsymbol{p = 0.2/100}$

Show all steps clearly.

Question 8: [6 marks]

Consider the following page reference string:

Page reference string: 3, 4, 1, 2, 4, 5, 0, 3, 6, 2

Ref String	3	4	1	2	4	5	0	3	6	2
f1										
f2										
f3										
H/PF										

Assume that there are **three frames**, and the **Optimal page replacement algorithm** page replacement algorithm is used. Answer the following questions:

a- Fill in the table above after copying it to your answer booklet. (5 marks)

b- Find the Page Fault Rate. (1 mark)

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Instructions:

- In the last **row**, you should indicate whether we have a hit (**H**) or a page fault (**PF**)
- **Tie-breaking rule:** In the Optimal algorithm, pages that will be used again in the future are not considered for tie-breaking. If multiple candidate pages either do not appear again in the future or are equally far in future use, replace the one that was **least recently used (LRU)** among them.

Question 9: [7 marks]

Assume having four processes P1, P2, P3 and P4 and the resources R1, R2, and R3 each having only 1 instance.

Below are the relations between the processes and resources:

- P1 → R1
- P2 → R2
- R1 → P2
- R2 → P3
- R3 → P1
- P3 → R3
- P4 → R1

a) Draw the corresponding **Resource-Allocation Graph** (RAG). (4 marks)
b) Convert the above RAG into a **wait-for** graph. (2 marks)
c) Determine if there is a deadlock and justify your answer. (1 mark)

End of questions

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